

Learning an Algebra of Distribution Shifts for Zero-Shot Compositional Generalization

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Abstract—Machine learning systems are routinely evaluated under individual distribution shifts, yet real deployments often encounter several shifts simultaneously. A model may have observed changes in illumination, sensor noise, demographic composition, or background during development while being tested under a previously unseen combination of these changes. Existing work on compositional generalization has established the importance of unseen combinations, particularly combinations of domains and classes, but most formulations treat the combination itself as the object to be generalized rather than learning an explicit representation of the *shift mechanism* that transforms one distribution into another. We propose Shift Algebra, a framework that represents distribution shifts as latent operators and learns to compose those operators. Given a reference distribution and controlled primitive interventions, the model learns a representation for each shift and an interaction-aware composition operator that predicts the representation and behavioral effect of an unseen combination. The framework explicitly distinguishes additive, synergistic, antagonistic, and order-dependent interactions. We further introduce ShiftCompose, a benchmark protocol in which primitive shifts and lower-order combinations are observed while selected higher-order combinations are held out. The central evaluation asks whether a model can predict the effect of an unseen shift composition without training directly on that composition. We define metrics for compositional generalization, interaction recovery, order sensitivity, and behavior prediction. The proposed formulation connects representation learning, robustness under distribution shift, compositional generalization, and controlled intervention-based evaluation. Importantly, this paper presents a research hypothesis and experimental protocol rather than fabricated empirical results; the experiments are specified so that the proposed claims can be tested reproducibly.

Index Terms—distribution shift, compositional generalization, out-of-distribution generalization, representation learning, robustness, intervention learning, zero-shot generalization, neural operators

I. INTRODUCTION

A central promise of machine learning is generalization: a model should perform well on examples that were not present during training. The difficulty becomes substantially greater when the data-generating distribution changes. Distribution shift is now a central concern in reliable machine learning, with research spanning domain generalization, robustness, covariate shift, subpopulation shift, corruption robustness, and out-of-distribution (OOD) generalization.

A particularly important but under-explored version of this problem occurs when multiple distribution shifts interact. Consider a classifier trained on images collected under several

conditions. During development it may encounter a low-light environment, a new camera sensor, and a change in background distribution, each separately or in some pairwise combinations. At deployment, however, the model may encounter all three simultaneously. The difficulty is not simply that the test distribution is “different.” Rather, the model has observed the *parts* of the change but not their particular composition.

This distinction matters because the number of possible combinations grows rapidly. If there are K primitive shift factors, the number of non-empty subsets is

$$2^K - 1. \quad (1)$$

Exhaustively collecting examples for every combination is therefore undesirable. A useful learning system should instead infer reusable structure about the shifts themselves.

Recent work has made compositional generalization under distributional shifts an explicit research problem. For example, compositional generalization has been studied for unseen domain–class combinations, and benchmark suites have been developed to measure this setting. These studies demonstrate that standard fine-tuning can struggle when familiar domains and familiar classes are recombined in previously unseen ways [1]. Other work studies compositional generalization across distributional shifts using neural-symbolic mechanisms [2]. These results establish that unseen combinations of known factors are important, but they do not by themselves answer the following question:

Can a model learn a reusable representation of a *distribution shift itself*, and compose such representations to predict an unseen higher-order shift?

We investigate this question through *Shift Algebra*. Instead of representing only the target domain, we represent the transformation that maps a reference distribution to the shifted distribution.

Let D_0 denote a reference distribution and let I_A denote a shift intervention. The shifted distribution is

$$D_A = I_A(D_0). \quad (2)$$

For two interventions A and B , a composition can be written as

$$D_{A \circ B} = I_B(I_A(D_0)). \quad (3)$$

The order is deliberately retained. In general,

$$I_B \circ I_A \neq I_A \circ I_B. \quad (4)$$

This non-commutativity is important because many real transformations interact: applying a compression transformation before adding noise need not produce the same distribution as applying noise before compression.

Our objective is not to assume that all shifts are additive. Instead, we learn whether and when a compositional structure exists. Each primitive shift is encoded into a latent shift representation, and a learned composition operator predicts the representation of an unseen composition. The model may also predict the effect of that composition on a trained predictor.

The paper makes four main contributions.

- 1) We formulate **shift composition** as an operator-learning problem in which the reusable object is a representation of a distribution transformation rather than only a representation of a domain.
- 2) We propose an **interaction-aware Shift Algebra** that separates primitive effects from interaction effects and explicitly permits non-commutative composition.
- 3) We introduce **ShiftCompose**, an evaluation protocol for zero-shot higher-order shift composition, where primitive shifts and selected lower-order combinations are observed while higher-order combinations are held out.
- 4) We define a set of tests that distinguish memorization of shift labels from genuine compositional generalization, including unseen triple shifts, reversed-order compositions, interaction recovery, and cross-domain transfer.

The intended contribution is therefore a research program at the intersection of representation learning, OOD robustness, and compositional generalization. We do not claim that compositional generalization or distribution-shift robustness is unexplored. Rather, the proposed novelty is the explicit treatment of *shift mechanisms as composable latent operators*, together with evaluation of whether learned operators can predict unseen higher-order shift effects.

II. RELATED WORK

A. Distribution Shift and Robust Generalization

Distribution shift occurs when the training distribution differs from the evaluation distribution. Classical formulations include covariate shift, label shift, and more general changes in the joint distribution. Modern work has established that even small or naturally occurring changes can materially affect model accuracy. ImageNetV2, for example, demonstrated that changes in data collection can produce measurable degradation even when the underlying task remains nominally the same [5].

Domain generalization studies learning from multiple source domains so that a model can generalize to unseen domains. Invariant learning approaches seek representations whose predictive relationships remain stable across environments [3]. Related work has examined the problem of recovering invariant feature subspaces under domain interventions [4].

These approaches generally ask whether a predictor remains stable under changes in the data distribution. Shift Algebra asks a different question: whether the *change itself* can be represented and composed.

B. Compositional Generalization

Compositional generalization studies the ability to construct correct behavior for novel combinations of familiar components. In language and structured prediction, this has long been associated with systematic generalization. More recently, compositional generalization has been studied directly under distributional shift.

Wang et al. introduced a formulation in which domain–class combinations can be held out while the individual domains and classes remain observed, and proposed Compositional Feature Alignment (CFA) to improve generalization to these unseen combinations [1]. Their work is particularly relevant because it demonstrates that compositional generalization across distributional changes is a distinct challenge rather than merely another domain-generalization benchmark.

Our setting differs in the object being composed. Instead of composing a domain with a class, we attempt to compose *transformations of a distribution*. Thus the primitive objects are shift operators such as illumination change, additive noise, style transformation, population reweighting, or controlled feature perturbation.

C. Compositional Models for Distributional Shifts

Conklin et al. investigated compositional generalization across distributional shifts using a differentiable tree machine, demonstrating that structured compositional mechanisms can improve systematic generalization [2]. This provides evidence that explicit compositional structure can be valuable under distributional changes.

Our proposed framework shares the broad motivation but differs in representation. We do not require a symbolic tree representation of the input or target task. Instead, we learn a continuous representation of the *shift transformation* and an operator for composing transformations. This permits the study of interaction strength, approximate commutativity, and prediction of model behavior.

D. Representation Learning and Latent Operators

Representation learning aims to map observations into latent spaces where relevant structure becomes easier to model. The proposed framework extends this principle from objects and examples to transformations.

Let

$$z_A = E(D_0, D_A) \quad (5)$$

be the latent representation of shift A . The central hypothesis is that there exists a composition function

$$z_{A \circ B} \approx C(z_A, z_B, \omega_{AB}), \quad (6)$$

where ω_{AB} captures interaction structure.

Unlike simple vector addition, C is not assumed to be commutative:

$$C(z_A, z_B) \neq C(z_B, z_A). \quad (7)$$

The model can therefore learn both approximately independent shifts and shifts whose order matters.

E. Gap in Existing Formulations

The literature establishes several important facts:

- 1) distribution shifts can significantly degrade model performance;
- 2) compositional generalization to unseen domain–class combinations is difficult;
- 3) explicit compositional structure can improve systematic generalization.

The proposed research investigates a more specific question:

Can distribution shifts themselves be represented as reusable operators whose composition predicts an unseen higher-order shift?

This distinction is important for the novelty claim. We do *not* claim that unseen combinations of distribution shifts have never been studied. Instead, the intended contribution is an operator-level representation, interaction algebra, and evaluation protocol focused on composing shift mechanisms.

III. PROBLEM FORMULATION

A. Reference Distribution

Let

$$D_0 = P_0(X, Y) \quad (8)$$

be the reference distribution. A model f_θ is trained on samples from D_0 or from a specified collection of source distributions.

A shift intervention I_A transforms samples or the data-generating process:

$$D_A = I_A(D_0). \quad (9)$$

Examples include:

- image brightness change,
- image corruption,
- background replacement,
- sensor noise,
- feature scaling,
- population reweighting,
- label imbalance,
- temporal drift,
- controlled changes to a nuisance variable.

The framework is agnostic to the exact intervention as long as the shift can be controlled or approximately characterized.

B. Primitive and Composite Shifts

For a set of primitive interventions

$$\mathcal{A} = \{A_1, A_2, \dots, A_K\}, \quad (10)$$

a composite intervention over subset S is

$$I_S = \prod_{i \in S}^{\rightarrow} I_{A_i}, \quad (11)$$

where the arrow indicates that the order may be relevant.

For example:

$$D_{A \circ B \circ C} = I_C(I_B(I_A(D_0))). \quad (12)$$

The main experimental condition is that D_A , D_B , D_C , and selected pairwise distributions are observed, while $D_{A \circ B \circ C}$ is held out during training.

C. Shift Representation

We define an encoder

$$z_A = E_\phi(D_0, D_A) \quad (13)$$

that maps a reference distribution and its shifted version to a latent representation of the transformation.

In a sample-based implementation, the encoder receives summary statistics or learned embeddings:

$$z_A = E_\phi(\text{Agg}(h_\psi(X_0)), \text{Agg}(h_\psi(X_A))), \quad (14)$$

where h_ψ is an optional feature encoder.

An alternative formulation operates directly on paired samples:

$$z_A = E_\phi(\{(x_i, I_A(x_i))\}_{i=1}^n). \quad (15)$$

The latter is useful when interventions are known at the sample level.

D. Composition Operator

Given primitive shift embeddings, a composition network predicts the embedding of a composite shift:

$$\hat{z}_{A \circ B} = C_\eta(z_A, z_B). \quad (16)$$

For a triple:

$$\hat{z}_{A \circ B \circ C} = C_\eta(C_\eta(z_A, z_B), z_C). \quad (17)$$

The model is trained on compositions available during training but evaluated on withheld compositions.

E. Interaction-Aware Algebra

Purely additive composition assumes

$$z_{A \circ B} = z_A + z_B. \quad (18)$$

This assumption is too restrictive. We therefore decompose the composite representation as

$$z_{A \circ B} = z_A + z_B + \gamma_{AB}, \quad (19)$$

where γ_{AB} is an interaction term.

For three shifts:

$$z_{A \circ B \circ C} = z_A + z_B + z_C + \gamma_{AB} + \gamma_{AC} + \gamma_{BC} + \gamma_{ABC}. \quad (20)$$

The higher-order term γ_{ABC} represents interaction that cannot be explained by pairwise effects alone.

This decomposition is not assumed to be the true causal decomposition of the data-generating process. It is a modeling hypothesis whose validity is tested empirically.

IV. PROPOSED METHOD

A. Overview

Shift Algebra contains four components:

- 1) a base representation encoder;
- 2) a shift encoder;
- 3) an interaction-aware composition module;
- 4) a behavior predictor that maps the predicted shift representation to model-level effects.

The overall pipeline is:

$$D_0, D_A \rightarrow z_A, \quad (21)$$

$$z_A, z_B \rightarrow \hat{z}_{A \circ B}, \quad (22)$$

$$\hat{z}_{A \circ B} \rightarrow \widehat{\Delta \mathcal{B}}, \quad (23)$$

where $\Delta \mathcal{B}$ denotes a measurable change in model behavior.

B. Feature-Space Shift Encoding

Let $f_\theta^{(l)}(x)$ denote the representation produced by a trained model at layer l . For a paired example $(x, I_A(x))$, define

$$\delta_A^{(l)}(x) = f_\theta^{(l)}(I_A(x)) - f_\theta^{(l)}(x). \quad (24)$$

The shift encoder aggregates these changes:

$$z_A = E_\phi \left(\{\delta_A^{(l)}(x_i)\}_{i=1}^n \right). \quad (25)$$

This representation has an important interpretation: it describes how the intervention changes the model's internal representation, rather than merely describing the shifted data.

C. Distribution-Space Encoding

To avoid dependence on one particular model, a second representation can be learned directly from the data distributions:

$$z_A = E_\phi(D_0, D_A). \quad (26)$$

A practical implementation may use distribution statistics such as:

$$\mu_A = \mathbb{E}_{x \sim D_A}[h(x)], \quad (27)$$

$$\Sigma_A = \text{Cov}_{x \sim D_A}[h(x)], \quad (28)$$

and higher-order learned statistics.

The final system can compare feature-space and distribution-space representations.

D. Order-Aware Composition

A standard multilayer perceptron can learn

$$\hat{z}_{A \circ B} = C_\eta(z_A, z_B). \quad (29)$$

To explicitly model order, we introduce an antisymmetric component:

$$C(z_A, z_B) = S(z_A, z_B) + \lambda R(z_A, z_B), \quad (30)$$

where

$$S(z_A, z_B) = S(z_B, z_A) \quad (31)$$

is the symmetric component and

$$R(z_A, z_B) = -R(z_B, z_A) \quad (32)$$

captures order-sensitive interaction.

One implementation is

$$R(z_A, z_B) = W(z_A \odot U z_B - z_B \odot U z_A), \quad (33)$$

with learnable matrices U, W .

The parameter λ controls the strength of the order-sensitive pathway.

E. Behavior Prediction

A central goal is not merely reconstructing an embedding but predicting the effect of an unseen shift on a trained model.

Let

$$\mathcal{B}(f, D) \quad (34)$$

denote a behavior vector containing metrics such as accuracy, calibration error, class-wise error, confidence statistics, or representation drift.

Define

$$\Delta \mathcal{B}_A = \mathcal{B}(f, D_A) - \mathcal{B}(f, D_0). \quad (35)$$

A behavior decoder predicts

$$\widehat{\Delta \mathcal{B}}_A = G_\omega(z_A). \quad (36)$$

For an unseen composition:

$$\widehat{\Delta \mathcal{B}}_{A \circ B \circ C} = G_\omega(\hat{z}_{A \circ B \circ C}). \quad (37)$$

This provides a stronger test than embedding reconstruction: the predicted shift representation must correspond to measurable changes in model behavior.

V. LEARNING OBJECTIVE

The total loss is

$$\mathcal{L} = \lambda_1 \mathcal{L}_{repr} + \lambda_2 \mathcal{L}_{comp} + \lambda_3 \mathcal{L}_{beh} + \lambda_4 \mathcal{L}_{order} + \lambda_5 \mathcal{L}_{reg}. \quad (38)$$

A. Representation Reconstruction

For observed composite shifts:

$$\mathcal{L}_{repr} = \|\hat{z}_{A \circ B} - z_{A \circ B}\|_2^2. \quad (39)$$

A cosine loss can be added:

$$\mathcal{L}_{cos} = 1 - \frac{\hat{z}^\top z}{\|\hat{z}\|_2 \|z\|_2}. \quad (40)$$

B. Composition Consistency

For equivalent decompositions:

$$A \circ (B \circ C) \quad (41)$$

and

$$(A \circ B) \circ C, \quad (42)$$

we can impose an associativity consistency objective:

$$\mathcal{L}_{assoc} = \|C(z_A, C(z_B, z_C)) - C(C(z_A, z_B), z_C)\|_2^2. \quad (43)$$

Importantly, associativity is a property of sequential composition itself and can be enforced only when the experimental intervention semantics support it. It should not be blindly imposed when the intervention representation contains stateful effects.

C. Behavior Loss

For measurable behavior:

$$\mathcal{L}_{beh} = \|G_\omega(\hat{z}_S) - \Delta\mathcal{B}_S\|_2^2. \quad (44)$$

For classification, this can include accuracy degradation, expected calibration error, and class-conditional error.

D. Order Loss

If both $A \circ B$ and $B \circ A$ are available:

$$\mathcal{L}_{order} = \|(\hat{z}_{A \circ B} - \hat{z}_{B \circ A}) - (z_{A \circ B} - z_{B \circ A})\|_2^2. \quad (45)$$

This prevents the model from collapsing all compositions into an unordered set.

VI. SHIFTCOMPOSE BENCHMARK

We propose a benchmark specifically designed to test the proposed hypothesis.

A. Benchmark Structure

A ShiftCompose task contains:

- 1) a reference distribution D_0 ;
- 2) a library of primitive interventions;
- 3) observed primitive shifts;
- 4) observed pairwise or low-order compositions;
- 5) held-out higher-order compositions;
- 6) a fixed prediction model whose behavior under those shifts can be measured.

The key rule is:

Every primitive shift used in a held-out composition must have appeared individually during training, but the complete composition must not have appeared.

For example, with primitive shifts $\{A, B, C, D\}$, the training set may contain

$$A, B, C, D, A \circ B, A \circ C, B \circ D \quad (46)$$

while the test set contains

$$A \circ B \circ C. \quad (47)$$

B. Synthetic Controlled Setting

A synthetic benchmark should be the first experiment because the true shift mechanisms are known.

For example, define

$$X = (X_1, X_2, X_3) \quad (48)$$

with

$$X_i \sim \mathcal{N}(\mu_i, \sigma_i^2). \quad (49)$$

Primitive shifts can independently modify:

$$A : \mu_1 \leftarrow \mu_1 + \alpha, \quad (50)$$

$$B : \sigma_2 \leftarrow \beta \sigma_2, \quad (51)$$

$$C : X_3 \leftarrow X_3 + \epsilon, \quad (52)$$

$$D : \text{Corr}(X_1, X_2) \leftarrow \rho'. \quad (53)$$

Controlled interaction can be introduced through

$$I_{A \circ B}(X) = I_B(I_A(X)). \quad (54)$$

This environment allows exact measurement of whether the model recovers known shift structure.

C. Vision Benchmark

A second benchmark can use MNIST, CIFAR-10, or a similar dataset.

Primitive transformations may include:

- brightness;
- contrast;
- Gaussian noise;
- blur;
- rotation;
- color shift;
- background transformation.

The experiment must ensure that the held-out composition is not accidentally represented in the training distribution.

For example, if the target is brightness + blur + noise, the training set may contain brightness + blur and brightness + noise but never all three together.

D. Natural Distribution-Shift Benchmark

A third benchmark should test whether the learned representation transfers beyond synthetic transformations.

Possible settings include:

- multiple image domains;
- demographic or subpopulation shifts;
- temporal shifts;
- sensor changes;
- cross-domain classification.

The challenge here is that primitive shifts are not perfectly controllable. Therefore, the benchmark should explicitly report uncertainty about shift identity rather than pretending that natural environments are clean interventions.

VII. EXPERIMENTAL PROTOCOL

A. Training Regimes

We define four regimes.

a) Regime I: Primitive-only training.: Only individual shifts are observed. The model must compose them without seeing any composite shift.

b) Regime II: Pairwise training.: Primitive and pairwise shifts are observed. Triple and higher-order compositions are held out.

c) Regime III: Sparse higher-order training.: Some triple shifts are observed while different triple combinations are held out.

d) Regime IV: Cross-domain transfer.: The model learns shift representations in one dataset and tests whether the same shift composition mechanism transfers to another dataset.

B. Baselines

The following baselines should be implemented.

- 1) **No-composition baseline**: directly predicts the target shift behavior from the target distribution features.
- 2) **Additive shift baseline**:

$$\hat{z}_{A \circ B} = z_A + z_B. \quad (55)$$

- 3) **MLP composition**: unconstrained MLP over concatenated shift embeddings.
- 4) **Attention composition**: cross-attention over primitive shift representations.
- 5) **CFA-style baseline**: a strong compositional feature-alignment method inspired by prior work on unseen domain-class combinations [1].
- 6) **Task-specific adaptation baseline**: retrain or adapt the base model directly on available shifted data.
- 7) **Oracle composition**: access to samples from the target composition, used only to establish an upper bound.

The additive baseline is particularly important because it tests whether the proposed interaction mechanism actually contributes beyond simple vector arithmetic.

VIII. EVALUATION METRICS

A. Compositional Generalization Gap

Let E_{seen} be the error on observed compositions and E_{unseen} the error on held-out compositions.

Define

$$CGG = E_{unseen} - E_{seen}. \quad (56)$$

Lower is better.

B. Relative Compositional Error

A normalized metric is

$$RCE = \frac{E_{unseen} - E_{seen}}{E_{oracle} - E_{seen} + \epsilon}. \quad (57)$$

This estimates how much of the gap to an oracle remains.

C. Behavior Prediction Error

For behavior vector $\Delta\mathcal{B}$:

$$BPE = \frac{1}{N} \sum_{i=1}^N \left\| \widehat{\Delta\mathcal{B}}_i - \Delta\mathcal{B}_i \right\|_2. \quad (58)$$

D. Interaction Recovery

For pairwise composition:

$$\hat{\gamma}_{AB} = \hat{z}_{A \circ B} - z_A - z_B. \quad (59)$$

Compare this with the observed interaction:

$$\gamma_{AB} = z_{A \circ B} - z_A - z_B. \quad (60)$$

The interaction recovery error is

$$IRE = \|\hat{\gamma}_{AB} - \gamma_{AB}\|_2. \quad (61)$$

E. Order Sensitivity

Define

$$OSI = \frac{\|\hat{z}_{A \circ B} - \hat{z}_{B \circ A}\|}{\|z_{A \circ B} - z_{B \circ A}\| + \epsilon}. \quad (62)$$

A model that always treats composition as commutative will fail this test whenever the ground truth is order-dependent.

IX. ABLATION STUDIES

A convincing paper should contain the following ablations.

A. Remove Interaction Terms

Set

$$\gamma_{AB} = 0. \quad (63)$$

If performance drops on strongly interacting shifts, this supports the interaction-aware formulation.

B. Remove Order Pathway

Remove the antisymmetric component R . This tests whether order modeling is useful.

C. Remove Behavior Supervision

Train only with representation reconstruction. This determines whether behavior supervision provides additional generalization.

D. No Pairwise Training

Hold out all pairwise compositions and test whether the model can compose from primitive shifts alone.

E. Different Latent Dimensions

Evaluate latent dimensions $d \in \{16, 32, 64, 128, 256\}$ to determine whether performance depends on a large latent representation.

F. Shift Encoder Capacity

Compare linear, MLP, attention-based, and set-based shift encoders.

X. THEORETICAL PERSPECTIVE

The framework can be interpreted as learning an approximate representation of a transformation semigroup.

Let $\mathcal{I}$ be a set of interventions with composition operation $\circ$. The exact intervention space satisfies

$$(I_A \circ I_B) \circ I_C = I_A \circ (I_B \circ I_C) \quad (64)$$

whenever the interventions are ordinary functions on the same state space.

We seek an embedding

$$\phi : \mathcal{I} \rightarrow \mathbb{R}^d \quad (65)$$

and a learned operation

$$C : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (66)$$

such that

$$\phi(I_A \circ I_B) \approx C(\phi(I_A), \phi(I_B)). \quad (67)$$

A successful representation therefore approximately preserves composition:

$$\phi(I_A \circ I_B) \approx C(\phi(I_A), \phi(I_B)). \quad (68)$$

This is analogous to learning a representation that respects an algebraic structure.

A. Commutator

The degree of non-commutativity can be quantified by

$$\kappa(A, B) = \|\phi(I_A \circ I_B) - \phi(I_B \circ I_A)\|. \quad (69)$$

If $\kappa(A, B) \approx 0$, the shifts approximately commute in the learned representation.

If $\kappa(A, B)$ is large, their order matters.

This produces a potentially interesting scientific object: a *shift interaction graph* whose edges encode how strongly pairs of distribution shifts interact.

B. Generalization Bound: Proposed Direction

A future theoretical analysis can assume that the true shift representation satisfies

$$\phi(I_A \circ I_B) = C^*(\phi(I_A), \phi(I_B)) + \epsilon_{AB}, \quad (70)$$

where ϵ_{AB} is composition approximation error.

For a composition of m shifts, if C is L -Lipschitz in each argument, recursive error propagation can be bounded by a function of L , m , and the primitive approximation errors.

A useful target theorem would be of the form

$$\|\hat{z}_S - z_S\| \leq \sum_{i=1}^{|S|} L^{|S|-i} \epsilon_i, \quad (71)$$

under suitable assumptions.

The practical significance is that the model's ability to generalize to higher-order compositions depends not only on representation quality but also on the stability of the learned composition operator.

A complete proof is left as future theoretical work and should not be claimed without formal verification.

XI. EXPECTED RESEARCH QUESTIONS

The project should be organized around the following questions.

- 1) **RQ1:** Can distribution shifts be represented in a latent space where composition is predictable?
- 2) **RQ2:** Does an explicit composition operator generalize to unseen higher-order combinations?
- 3) **RQ3:** Do interaction-aware operators outperform additive composition?
- 4) **RQ4:** Can the learned representation predict the effect of an unseen shift on model behavior?
- 5) **RQ5:** Can the representation detect whether two shifts approximately commute?
- 6) **RQ6:** Does a shift representation learned on one dataset transfer to another dataset?

XII. EXPECTED OUTCOMES AND FALSIFICATION CRITERIA

This section is intentionally phrased as hypotheses rather than results.

A. Hypothesis H1

Models with explicit shift composition should have a smaller compositional generalization gap than models that directly learn each observed distribution.

B. Hypothesis H2

Interaction-aware composition should outperform purely additive composition when primitive shifts have non-linear interactions.

C. Hypothesis H3

Order-aware composition should improve performance on non-commutative transformations.

D. Hypothesis H4

A shift representation trained with behavior supervision should predict unseen model degradation more accurately than a representation trained only to reconstruct shifted distributions.

The project should be considered unsuccessful if:

- additive composition performs as well as the proposed method across all datasets;
- no consistent benefit exists on unseen compositions;
- the learned latent space cannot predict model behavior;
- performance depends primarily on memorizing shift labels;
- order sensitivity is not recoverable when the underlying transformations are non-commutative.

These failure criteria are important because they prevent the paper from becoming a method proposal that assumes its own success.

XIII. WHY THIS PROBLEM MATTERS

The practical motivation is straightforward. If a system has K possible shift factors, exhaustive testing of every combination requires potentially exponential data collection.

A compositional representation offers a different strategy:

learn primitive effects $\rightarrow$ learn interactions $\rightarrow$ predict unseen combinations (72)

The potential benefit is not limited to one application area. The same principle could apply to:

- computer vision under combined corruptions;
- speech recognition under speaker, noise, and channel shifts;
- language models under domain and style changes;
- medical ML under demographic and acquisition shifts;
- forecasting under temporal and environmental shifts;
- reinforcement learning under changes in environment dynamics.

The key scientific question is whether these shifts have reusable structure that can be learned rather than treated as unrelated OOD events.

TABLE I

ZERO-SHOT COMPOSITIONAL GENERALIZATION ON HELD-OUT SHIFT COMBINATIONS. LOWER CGG AND BPE ARE BETTER; HIGHER ACCURACY IS BETTER. VALUES SHOULD BE FILLED ONLY AFTER EXPERIMENTS ARE COMPLETED.

Method	Accuracy	CGG ↓	BPE ↓
No composition	—	—	—
Additive	—	—	—
MLP composition	—	—	—
Attention composition	—	—	—
CFA-style baseline	—	—	—
Shift Algebra	—	—	—
Oracle	—	—	—

XIV. LIMITATIONS

The proposed formulation has several limitations.

First, controlled interventions are easier to study than naturally occurring shifts. Real-world distribution changes may not decompose cleanly into identifiable primitive interventions.

Second, the latent representation is not automatically causal. A representation that predicts shift behavior does not prove that it recovered the true mechanism responsible for the shift.

Third, composition may be state-dependent. Applying B after A may change the effect of B , so a fixed operator representation may be insufficient. A richer formulation would use context-dependent operators:

$$z_{B|A} = C(z_B, z_A). \quad (73)$$

Fourth, the benchmark can be unintentionally contaminated. If the training distribution contains enough examples that approximate the held-out combination, the zero-shot claim becomes weaker.

Fifth, the number of possible higher-order combinations grows exponentially. The proposed approach is therefore useful only if it can generalize substantially beyond the compositions explicitly observed during training.

XV. REPRODUCIBILITY PLAN

All experiments should use fixed random seeds and predefined train/validation/test intervention sets.

For every benchmark, release:

- 1) the primitive intervention definitions;
- 2) the composition graph;
- 3) the exact held-out combinations;
- 4) random seeds;
- 5) model configurations;
- 6) training checkpoints;
- 7) evaluation scripts;
- 8) per-shift and per-composition metrics.

The benchmark split must be generated before model development and kept fixed to avoid adaptive test-set selection.

XVI. PROPOSED EXPERIMENTAL TABLE

The final paper should report a table of the following form.

No numerical result should be inserted into this manuscript until it has been measured.

XVII. DISCUSSION

The main conceptual shift in this work is to treat a distribution shift as a learnable object.

Traditional OOD evaluation asks:

How does the model perform on D_{test} ?

Shift Algebra asks:

What transformation took us from D_0 to D_{test} , and can that transformation be composed from transformations already observed?

This reframing may allow distribution-shift research to move from a catalog of independent robustness failures toward a structured theory of how failures and behavioral changes compose.

The idea also connects naturally to representation learning. A useful representation is not merely one that compresses observations; it should preserve the structure relevant to downstream reasoning. If the structure of distribution shifts is compositional, then a representation that preserves composition could provide a principled route to generalization.

The most interesting outcome would therefore not simply be a few percentage points of improved accuracy. A stronger scientific result would demonstrate that the latent shift space exhibits measurable algebraic properties: approximately additive regions, strongly interacting regions, and non-commutative pairs. Such a result would provide a new language for describing distributional change.

XVIII. CONCLUSION

We proposed Shift Algebra, a machine learning framework for representing distribution shifts as composable latent operators. The central hypothesis is that models can learn reusable representations of primitive distribution transformations and use them to predict unseen higher-order combinations.

The proposed framework differs from ordinary domain generalization by making the transformation itself the object of representation. It differs from standard compositional generalization by composing shift mechanisms rather than only domains, classes, or symbolic structures. It further introduces interaction and order sensitivity so that the model does not assume all distribution shifts combine independently.

We proposed ShiftCompose, a benchmark protocol in which primitive shifts and selected lower-order compositions are observed while higher-order compositions are held out. We also proposed metrics for compositional generalization, behavior prediction, interaction recovery, and non-commutativity.

The central empirical question is ultimately falsifiable:

Can a model predict the effect of a distribution shift it has never seen, f

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If the answer is yes, the result would suggest that distribution shifts possess learnable reusable structure. If the answer is no, the failure would still clarify an important limitation of compositional approaches to robustness.

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This manuscript is a research proposal and paper draft. Experimental claims, numerical results, and theoretical guarantees should be added only after the corresponding experiments and proofs have been completed.

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